

Commit 0d30bde

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ActuallyNull committed on Aug 11

model v1.1; added earlystopping and reducelronplateaucallback

main1 parent 6e25aff commit 0d30bde

Filter files...

- data_preperation.py
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2 files changed +15 -5 lines changed

Search within code

data_preperation.py-4

@@ -1,11 +1,7 @@

1 1 from data_processing import preprocess_dataset

2 - import numpy as np

3 2 from sklearn.preprocessing import LabelEncoder

4 3 from sklearn.model_selection import train_test_split

5 4

6 -

7 -

8 -

9 5 def create_train_val_test_splits():

10 6

11 7 X = []

model.py+15 -1

@@ -1,10 +1,17 @@

1 1 import tensorflow as tf

2 2 from keras import layers

3 + from keras.callbacks import EarlyStopping, ReduceLRonPlateau

3 4 from data_preperation import create_train_val_test_splits

4 5 from sklearn.metrics import confusion_matrix, classification_report

5 6

6 7 X_train, y_train, X_val, y_val, X_test, y_test = create_train_val_test_splits()

7 8

9 + callback = [

10 + EarlyStopping(monitor='val_loss', patience=7, verbose=1, restore_best_weights=True),

11 + ReduceLRonPlateau(monitor='val_loss', factor=0.2, patience=5, min_lr=1e-6, verbose=1)

12 +]

13 +

14 +

8 15 model = tf.keras.Sequential([

9 16 layers.Conv1D(32, kernel_size=5),

10 17 layers.Conv1D(32, kernel_size=5),

@@ -33,7 +40,14 @@

33 40 loss='sparse_categorical_crossentropy',

34 41 metrics=['accuracy'])

35 42

36 - model.fit(X_train, y_train, epochs=10, batch_size=32)

43 + model.fit(X_train,

44 + y_train,

45 + epochs=100,

46 + batch_size=64,

47 + callbacks=callback,

48 +)

49 +

50 + print(model.summary)

37 51

38 52 test_loss, test_acc = model.evaluate(X_test, y_test)

39 53 print(f"Test accuracy: {test_acc:.4f}")

Comments 0

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